TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Developing play tarot cards to support playful learning in teacher education

Citation for published version (APA):
Skovbjerg, H. M., Jørgensen, H. H., Pérez Quiñones, K. Z., & Bekker, T. (2022). Developing play tarot cards to support playful learning in teacher education. In *DRS2022: Research papers* Article 187 (Proceedings of DRS; Vol. 2022). Design Research Society. https://doi.org/10.21606/drs.2022.187

Document license:
CC BY-NC

DOI:
10.21606/drs.2022.187

Document status and date:
Published: 01/01/2022

Document Version:
Publisher's PDF, also known as Version of Record (includes final page, issue and volume numbers)

Please check the document version of this publication:

• A submitted manuscript is the version of the article upon submission and before peer-review. There can be important differences between the submitted version and the official published version of record. People interested in the research are advised to contact the author for the final version of the publication, or visit the DOI to the publisher's website.
• The final author version and the galley proof are versions of the publication after peer review.
• The final published version features the final layout of the paper including the volume, issue and page numbers.

Link to publication

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

• Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
• You may not further distribute the material or use it for any profit-making activity or commercial gain
• You may freely distribute the URL identifying the publication in the public portal.

If the publication is distributed under the terms of Article 25fa of the Dutch Copyright Act, indicated by the "Taverne" license above, please follow below link for the End User Agreement:

www.tue.nl/taverne

Take down policy
If you believe that this document breaches copyright please contact us at:

openaccess@tue.nl

providing details and we will investigate your claim.

Download date: 07. Nov. 2026

Design Research Society
DRS Digital Library

DRS Biennial Conference Series DRS2022: Bilbao

Jun 25th, 9:00 AM

Developing play tarot cards to support playful learning in teacher education

Helle Marie Skovbjerg
Design School Kolding, Denmark

Helle Hovgaard Jørgensen
UC Lillebaelt

Keila Zarl Perez Quinones
Design School Kolding, Denmark

Tilde Bekker
Eindhoven Technical University

Follow this and additional works at: https://dl.designresearchsociety.org/drs-conference-papers

 Part of the Art and Design Commons

Citation

Skovbjerg, H.M., Jørgensen, H.H., Perez Quinones, K.Z., and Bekker, T. (2022) Developing play tarot cards to support playful learning in teacher education, in Lockton, D., Lenzi, S., Hekkert, P., Oak, A., Sádaba, J., Lloyd, P. (eds.), *DRS2022: Bilbao*, 25 June - 3 July, Bilbao, Spain. https://doi.org/10.21606/drs.2022.187

This Research Paper is brought to you for free and open access by the DRS Conference Proceedings at DRS Digital Library. It has been accepted for inclusion in DRS Biennial Conference Series by an authorized administrator of DRS Digital Library. For more information, please contact dl@designresearchsociety.org.

DRS 2022
Bilbao 25th June — 3rd July

Developing play tarot cards to support playful learning in teacher education

Helle Marie Skovbjerg[a*], Helle Hovgaard Jørgensen[b] , Keila Zari Pérez Quiñones[a], Tilde Bekker[c]

[a]Design School Kolding, LAB Design for Play, Denmark

[b]University College Lillebælt, Denmark

[c]Eindhoven Technical University, Netherlands

*Corresponding author email: skovbjerg@skovbjerg.dk

https://doi.org/10.21606/drs.2022.187

Abstract: Via two design-based research experiments, this paper explores how we can design a tool to support designing for playful learning in teacher education. Several recent review studies show that integrating play qualities into a learning context is not always easy. We design a set of tarot cards with the aim of exploring actions in learning situations and play qualities for those specific actions. Our experiments show that using the tarot cards as a way for students and teachers to reflect and come up with further playful learning designs brings in a broader diversity of play qualities, especially qualities that are not commonly seen as productive in an educational context.

Keywords: play design; playful learning; education; design research; design-based research

1. Introduction

Several recent studies of teaching in higher education have shown that playful approaches to teaching are growing (Moseley & Whitton, 2019; Nørgård et al., 2017; Whitton, 2018). These studies indicate that playful approaches can meet some of the challenges that future education faces. The education of the future call for being able to attract skilled and committed young people to teaching professions; they clarify that the professions must remain relevant, among other things, by creating a close connection between education and profession, and in the long run, the professions can help to ensure that children and young people continue to be engaged in learning and development. Playful approaches focus on the process; they view learning processes as simultaneously social and an exercise in standing in the open, overcoming uncertainty and having faith in the power of others' and personal creativity. As Nørgård et al. (2017) point out, 'Playful learning provides a new paradigm for understanding higher education pedagogy in an increasingly performative risk-aversive environment. It is an approach that gives learners and teachers freedom to be

This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International Licence.

playful, freedom to make choices, and freedom towards the world' (p. 276). At the same time, the hope is that future teachers will be equipped to practice their profession among children and thus also base their interaction with and teaching of children on the same values.

As part of the research project *Playful Learning Research Extension*[1]*,* review studies have shown that there are several challenges associated with working with playful approaches to teaching (Boysen et al., 2022; Jensen et al., 2021; Jørgensen et al., 2022). In particular, the interconnection between play and teaching is controversial and contradictory, both theoretically and practically. Theoretically, play is challenging to deal with (Skovbjerg, 2020, 2021; Skovbjerg et al., 2021; Sutton-Smith, 2001), and it is concretely expressed by a large amount of ambiguity in the manner in which the concept of play is presented. Diffuse ideas dominate, and the language and concepts challenge us. At the same time, the notion of 'free play', as something that must be protected against learning, is strong; this makes it difficult to work with playful learning in a meaningful way (Jørgensen & Skovbjerg, 2021; Jørgensen & Skovbjerg, 2020; Øksnes & Sundsdal, 2020). The use of the term 'playing' also means that it clashes with the efforts in educational contexts where learning is foregrounded, and the contrast between play and learning is constantly evoked (Pramling & Johansson, 2006; Skovbjerg, 2021). As Stenros (2015) states, 'It is problematic that there is an ideological stance found in much of the literature on serious games and gamification that posits that games and play are somehow inferior unless they are useful' (p. 147). There is a need to explore how to design relevant knowledge that can be used to overcome the ambiguity of the play concepts and the discrepancy between playful attitudes and educational purposes. Based on several design experiments within the tradition of design-based research, this paper explores the following design research question: *How can we develop a tool that supports playful learning in teacher education?*

We were inspired by the quality of tarot cards as a system that uses open-ended visualisations as triggers for diverse interpretations (Pollack, 2002) to create a tool for presenting and interpreting playful learning knowledge. The history of tarot cards goes back to the 15th century when the tarot was originally created as a deck of cards for playing games. The cards' occult symbolism as we understand it today was not associated with the cards until the late 18th century, when tarot cards were repurposed with specific meanings for fortune-telling. They consist of 22–56 cards divided into four houses, with visual symbols from Christianity, the Jewish kabbalah, Greek mythology and Egyptian mythology; there are also references to numerology, astrology and the herbaria alphabet. These archetypal images (Semetsky, 2011) give people a rich opportunity to use their imagination and associations to open conversations. Tarot cards are grounded in Carl Gustaf Jung's psychology, where the visuals invite people into a conversation about their real-life

[1] Playful Learning Research Extension is a research project funded by the LEGO Foundation with the aim of exploring combining playfulness and learning in higher education and among education teachers. Six university colleges in Denmark are involved, including 12 PhD students and 8 senior researchers, as well as collaborations with three Danish Universities.

individual experiences (Auger & Banzhaf, 2000; Hamaker-Zondag, 1997; Lindley et al., 2020). Often, a person uses the cards for another person, performing a reading as a one-to-one conversation; the cardholder starts with three to six cards, placing them face down on a table and revealing them as the conversation progresses. The cardholder asks the person to choose a card; the card is turned over, and the person being interviewed is asked questions.

In our work, we are inspired by interaction design and the idea of lenses of play presented as a card tool (Bekker et al., 2014); based on Ericson, Bekker et al. emphasise the importance of being able to shift perspective when being concerned about play. As Bekker et al. (2014) state, "The purpose of the lenses is to support the design and to be able to shift between perspectives when exploring different design directions and decisions. Just like a sculptor who uses a lens to look at a detail of his stature, and then looks at the overall picture again, the lens helps to temporarily focus on a specific detail, without losing sight of the broader picture" (p. 264). The different lenses of play in the present paper are explored through the play mood perspective developed by Skovbjerg (2021). This play perspective focuses on play as a set of actions that make the participants able to enact different situations that are experienced as having play qualities—described as being *in* a play mood. From the perspective of Skovbjerg (2021) and inspired by Lucero and Arrasvuori (2013), play is closely related to actions. This understanding concurrently supports a learning perspective as described by Dewey (1986) and Lennon (2015), where knowledge and learning processes are seen as something learners do to create a shared world, where we can experience learning insights.

The purpose of our research is to explore how to develop a tangible playful learning tarot card deck with the aim of supporting designing activities for playful learning. The main contributions of the paper are, on the one hand, to show the importance of translating ambivalent play knowledge into concrete designs, where it is possible to act upon them, and on the other hand, to show how a tool can be used to support creating future playful learning designs within the education of teaching professionals.

2. Research context and method

This paper is based on two design experiments created in connection with the research project Playful Learning Research Extension. The research project (2019–2023) is funded by the LEGO Foundation and involved researchers from six university colleges educating teachers in Denmark, Danish universities and a number of foreign partners.

The research project stems from design-based research (Barab & Squire, 2004; Brown, 1992; Ejsing-Duun & Skovbjerg, 2019), which is characterised by close collaboration with a field of practice—in this case, teacher education. This means that a number of design experiments are initiated in teacher education. Considering the local educational practice and need, we are developing methods that make it possible to create both new forms of teaching and playful learning and systematically investigate them. The design of the play tarot cards helps

us generate knowledge, while at the same time, the knowledge must be relevant to an educational field of playful learning. As part of the process, we continuously explore and generate theory that will form the basis of understanding for playful learning in a lifelong context, and as Barab and Squire (2004) prescribe in their case, we focus on 'the messiness of real-world practices' (p. 3). This means that flexible revisions of the design are being tried out and the participants are taking an active part in the experiment and helping to ensure documentation in the form of filling in templates and creating padlets, presentations and publications that can be the subject of further analyses. With design-based research, we are preoccupied with what characterises concrete situations, where playful learning and teaching take place and how theory development can both make it easier to understand these and create opportunities to initiate and thus design future teaching situations. Design-based research enables both a prescriptive and a descriptive dimension (Ejsing-Duun & Skovbjerg, 2019).

The design process was inspired by models from Christensen et al. (2012) and Barab and Squire (2004) that include four phases, and the development of the cards is structured according to the model (see Figure 1). In the **context domain**, we investigated how playful learning is thematised and used both theoretically and practically. In the **lab domain**, we formulated concrete principles for what we wanted to explore further through concrete actions, in combination with the theory of play and needs that we had uncovered in the context phase. In the **experiment domain**, we experimented with our ideas through concrete design experiments. Finally, in the **reflection domain**, we looked at the empirical material and reflected on what the playful learning was based on, and more specifically, what the cards could bring forward into the next iteration.

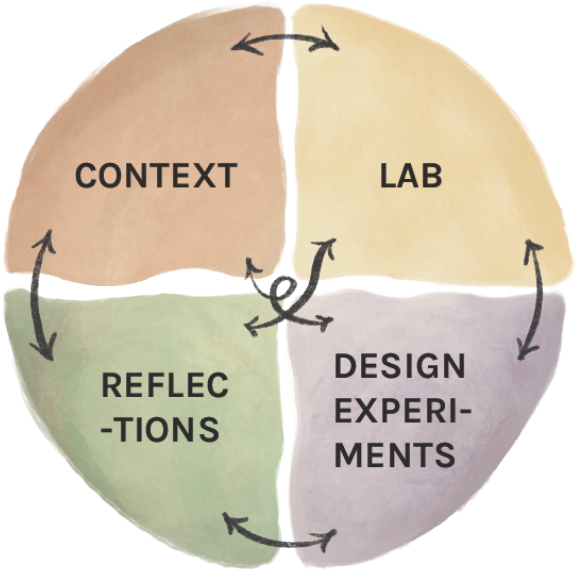


Figure 1: The Design-based research model

The design experiments were documented via field notes, field photos and sensory ethnography (Pink, 2001), as well as concrete productions, such as padlets, didactic designs and presentations. In this paper, we present two design experiments with the cards that involved the four phases described in the model. The two design experiments were performed to develop the play tarot cards iteratively.

The analysis strategy was inspired by grounded theory and situational analysis (Charmaz, 2014; Clarke, 2011). We mapped each experiment and coded using open coding. Furthermore, we used the insights to provide direction for the next experiment.

3. Theoretical framework for developing the cards

The theoretical framework for designing the tarot deck was based on a concept of playful learning being closely related to actions on one hand (Dewey, 1986; Lennon, 2015), and on the other hand, involving those actions being experienced as having play qualities (Skovbjerg, 2021; Skovbjerg & Jørgensen, 2021). Learning is about doing something in the real world, where the disciplines within teacher education are enacted through experiences and through actions that we experience as a shared world. We understand the shared world as a dialogical imaginative world (Bakhtin, 1981; Lennon, 2015). Students can experience the actions we design for them to be involved in as having play qualities (Skovbjerg, 2021; Skovbjerg & Jørgensen, 2021). Such actions are often driven by the participants; in line with Bakhtin, they can be joyful and meaningful to those who participate; and they indicate a social learning approach.

3.1 Play actions indicate play qualities

Based on an extensive ethnographic study of children's play behaviour Skovbjerg (2021) presents four different basic types of actions with play qualities, pointing to four basic play moods. These actions are as follows:

1) S*liding,* where the aim is making as little change as possible from action to action. Imagine a play situation playing with LEGO, building with one brick, then another brick. One action follows the next one, with as little change as possible. The play quality of sliding, experienced as devotion mood, includes flow and continuity, where conflict or resistance is not part of the play quality.
2) *Shifting*, can be seen as a wild roller coaster trip. The actor starts by going slowly up, getting ready for a change in speed and height. Suddenly, the direction changes and the fast movements create a feeling of unpredictability. The strong repetition is a play quality, but the acts of play change through the changes in height, speed and direction. The play quality of shifting is the unpredictable change, using the body as a tool for the change and opening up for other experiences. The mood experience is described as intensity.

3) *Displaying,* where actions are often related to performing or putting oneself and/or one's skills on stage. Dancing or singing, taking photos of others and dramatic role-playing are examples of acts of displaying. The focus is on showing off and making oneself into an object of evaluation. Display involves not only following others but also being able to make the action 'swing', which means being able to make changes from one beat to the next, by including personal interpretation. The quality of play is related to the expectation of change and the mood is described as experiencing tension; therefore, displaying has a weaker beat compared with the sliding act and devotion mood. The players expect to present a personal style and thus a change from action to action.
4) *Exceeding*, where acts are fleeting and the players expect the doing of play to exceed and change over time, again and again. It is anticipated that the play-act will be 'out of beat' rather than finding a rhythm. Bizarre jokes, jackass tricks, dirty words and stories of frivolity are characteristics of play acts that exceed. The experience of play quality is described as a euphoric mood. Play culture is hereby linked to folk culture and is comparable to Bakhtin's (1984) descriptions of carnival culture: 'A boundless world of humorous forms and manifestations opposed the official and serious tone of mediaeval ecclesiastical and feudal culture. In spite of their variety, folk festivities of the carnival type, the comic rites and cults, the clowns and fools, giants, dwarfs and jugglers, the vast and manifold literature of parody - all these forms have one style in common: they belong to one culture of folk carnival humour' (p. 4).



Figure 2: Examples of the designed tarot cards, each from one of the four main play-action groups related to the four mood groups (see also Table 1).

5. Prototyping the tarot cards as a point of departure

5.1 An existing play universe with aesthetic meanings

To create the first prototype of the cards, we employed the perspectives of playful learning and play qualities (Dewey, 1986; Lennon, 2015; Skovbjerg, 2021). We were inspired by the tarot card as a design system to develop a deck of sixteen cards, based on the four types of play actions previously described (see Table 1). From the historic knowledge about tarot, we borrowed the original intention of use as a play tool. Other card tools do not necessarily invite participants into playful experiences while using the cards. We wanted the cards themselves to have playful qualities. Tarot cards traditionally rely on characters with carnivalesque features, which we also incorporated into our designs. All the cards we illustrated include a direct actor (subject or character) that the conversation can be addressed to or around. We also appropriated the modern association tarot cards have with fortune-telling. Although we do not draw on the intention of fortune-telling (the act of predicting someone's or something's future), we utilise the storytelling and inquisitive nature of that act. We use tarot as a play universe we can buy into as adults (Skovbjerg, 2021). The aesthetic choices help the user build narratives because they cue a language and a tone we can recognise and imitate to carry out a play experience. Much like play universes created for children, which can rely on characters and stories that the children then take and create independently, we collect notions of playfulness to create possibilities. We do this by referencing images of circus characters, carnivalesque aesthetics, rituals, mystery, magical illusions and theatrics. In combination, these characteristics form a system of design affordances in which the user can identify images and a structure that facilitates a path for a playful attitude to arise while giving the user an idea of how to make use of them and how to 'behave' with them. When drawing on the carnivalesque culture and connecting it to children's play, the design choices are aligned with ideas from Bakhtin's theory of the connections between folk culture, carnival and language. The carnival is about 'the unmasking and disclosing of the unvarnished truth under the veil of false claims and arbitrary ranks [...] Bakhtin repeatedly points to the Socratic dialogue as a prototype of the discursive mechanism for revealing the truth' (Bakhtin, 1981, p. 426). In other words, by drawing on the carnivalesque, we use an expressive and imaginative figurativeness that foregrounds a dialogue that is revealing, open and equal.

5.2 Prototype design choices

The aesthetic expression of the cards can then be described as a bricolage of the material at hand, both borrowed visuals and meanings of existing play universes. To describe this more concretely through our physical prototype, we detail some of the design choices made. First, based on the four basic play actions and previous ethnographic data, we identified a number of sub-actions, four for each main play action, which we could use as inspiration for the prototype (see Table 1). These four main categories construct our play tarot card suits inspired by the traditional tarot cards (swords, pentacles, cups and wands). The four play

tarot suits we subsequently create are the houses of Devotion (related to the act of *sliding*), Intensity (related to the act of *shifting*), Tension (related to the act of *displaying*) and Euphoria (related to the act of *exceeding*). Each card is then visually constructed of the following three elements: At the top is the initial letter of the house the card corresponds to (D, I, T, E), followed by a captivating image collage with the main actor; on the bottom is the title of the card (see Figure 2). The title is constructed by a participle phrase that highlights the action verbs carried out by the actor. Some examples of titles are *'The builder is building'*, *'The swinger is swinging'* and *'The collector is collecting'*. This repetition is intentional and aims to highlight actions that are all drivers for play by having both an actor who does something and the action they do. In addition to the focus on actions, these choices of wording carry a humorous, playful attitude by making what is said slightly redundant. The illustration, which is evidently the most noticeable element, is characterised by a collage of black and white images and colourful illustrations of both abstract and recognisable shapes. The composition follows a few rules, none of which the participants making use of the cards need to be verbally warned of prior to the activity, but we believe to have implications in the effective visual communication of the tool. The two main rules related to the composition are that the actor of the card is illustrated by a somewhat recognisable character/subject placed in the centre, and this needs to be a black and white image. Examples of these actors are a female circus juggler, a pair of hands and a yelling girl. The consideration behind having this main actor be extracted from a realistic photograph is to provoke an immediate interpretation in the viewer. Once that actor is recognised, the viewer can continue to take in the rest of the visual information, which is more abstract and imaginary. Altogether, the visual is intended to induce the viewer into a mode of creative thinking. Finally, the letter that represents the play suit or house is perhaps the element adding the most complexity to the cards in the sense that it requires the facilitator to have at least a basic understanding of a specific play theory (Skovbjerg, 2021) but has the least visual presence. For this first round of experiments, we intentionally focus on how well the combination of image and title alone could facilitate the use of the tool to describe enacted play experiences; as yet, we do not give much attention to how the theoretical layer added by having previous knowledge of the four houses can add more dimensions to our play tarot to be used as a design tool.



*Figure 3: Card for 'performing' corresponding to the **displaying** play actions group (see Table 1). The performer is performing, from the House of Tension.*

Table 1: Overview of ground actions and moods, related play quality actions

Ground action/ mood	**Sliding/ devotion**	**Shifting/ intensity**	**Displaying/ tension**	**Exceeding/ euphoria**
	Building	Balancing	Spectating	Smashing
	Fiddling	Jumping	Performing	Yelling
	Collecting	Running	Pretending	Destroying
	Imitating	Swinging	Dancing	Mocking



Figure 4: Examples of the tarot cards

6. Design experiments with the prototype of the tarot cards

6.1 Design experiment with teachers in teacher education

Based on the **context domain**, where we learned about the difficulties related to play qualities in the context of playful learning, in the **lab domain**, we conceptualised our tarot card tool and the first experiment. For this first trial, which constituted the initial part of our **experiment domain,** we invited 35 teachers from teacher education to participate in a design experiment trying out the set of 16 play tarot cards. We designed the experiment with inspiration from traditional tarot reading examples from a performative angle. The facilitator/reader invited a person into the tarot conversation with the aim of exploring experiences of play qualities while the other teachers observed and worked as a reflecting team. The situation included a black tablecloth of velour, candles and crystals to set the scene and draw on the atmosphere of mystery and occultism. We placed four randomly chosen cards on the table, facing down. Inspired by the episodic interview, where concrete actions and situations are explored in a sensory manner pointing to emotions and experiences (Flick, 2001), we asked the person to start thinking about a specific learning situation where the person experienced play qualities because of that specific learning situation. We asked the person to elaborate on the following questions:

- Where were you?
- What did you do?
- What materials did you use?
- What happened?

The questions all focus on actions—on what the person did.

After having a sensory feeling of the concrete episode described thoroughly by the person as guided by our questions, we revealed the first card by flipping it over to explore the situation further. The revealed card was then used to elaborate on the learning experience through play qualities, particularly the one the card was suggesting. We flipped over the other three cards, and we ended up using eight randomly selected cards in total to explore that specific situation through the different play actions and qualities. When we had explored the interpretation of a specific learning situation with experiential play qualities, we invited the person to choose two cards from the full deck that the person wanted to incorporate in a future playful learning design. We will illustrate the procedure with the following scenario: She chose The Smasher and The Yeller. The participant elaborated further on the experiences while looking at the two chosen cards: *"The one about yelling, I find interesting. I see…often it is very polite in the learning activities which I am designing for. There is something about that voice*". In the following the interaction with the cards make her realise something important about her teaching practice:

> When I teach, and the students are creating something, I often become really disappointed that the students do not care about their creations. They throw it away without me seeing it. I would like them to bring it to their exam, but it might be that we should do the destruction of the creations instead. Or maybe burn it up […] I realise now that I am very much in my head when I design, instead of thinking about the sensory quality.

The participant chose the cards *The Smasher is smashing* and *The Yeller is yelling* because these cards made the participant realise two things. First, smashing and yelling are play qualities. When in play, it is meaningful to smash and yell; the participant realised this while looking and pointing her finger at the illustrations of The Smasher and The Yeller. These illustrations have a comic feel that appears via the exaggerated expression of a yelling girl on the phone and a cake-on-the-face smashing scene. Second, inspired by the cards, the participant realised that those two qualities could provide a play solution to their problem. Instead of creating an awkward situation for both the participant and students, it was possible to make a scene of engaging in an exceeding act of yelling and smashing their own creations. The cards provoked the following reflection on the situation:

> [T]hey [the students] have made up their minds. We might as well smash it [their creations]. What would happen...? It might even set free some energy. I will try it.

The play tarot universe and the interview focus on exploring play qualities through actions. The users are challenged to creative thinking and reflections on play qualities closely related to the specific learning situations. The cards seem to have the potential to nudge participants to foresee future playful learning situations, inspiring them to apply play qualities that they do not usually apply.

The main insights from the first experiment were framed in the **reflection** domain: the tarot cards had an appeal because of both their materiality and their special figurative design and

aesthetics. Our participants pointed at them and took them in their hands to investigate the pictures; they were clearly inspired by the visuals on the cards. We learned that the design of play qualities in a diverse sense, not relating to cultural codes specific to education, seem to invite teachers into another type of conversation about what playful learning designs could be. This could indicate that the cards might be able to help teachers in the transformation of their thinking about their teaching.

6.2 Design experiment with students

For the next experiment, we wanted to explore the material, the transformative possibility and the aesthetic qualities further defined in the **context domain**. In the **lab domain**, we carried out playful learning in two classes of first-year teacher students (all in their 20'ies). The subject was in both cases, Danish & Technology, and we planned for playful learning by involving Ozobots (small robots on wheels) in storytelling. In groups of three to five students, they invented stories performed by the Ozobots. We initiated both sessions by asking for volunteers to participate in the tarot interview after class. During class, we moved around the groups discussing ideas, storylines and presentations of their Ozobot-story. At the same time, we organised informed consent forms in relation to conducting a video ethnography of the activity (Pink, 2001). While moving around the groups we succeeded in finding a group of students for the subsequent interview. We split the group of students in two and carried out two interviews. In this paper, we report on one of the interviews with three of the students.

In the **experiment domain**, the researcher initiated the tarot interview by saying, 'Now we want to investigate the play qualities of the situation'; the researcher then gave a short description of what was meant by play qualities. The three students were asked to draw two cards each. They expressed surprise when seeing the figurative design of the cards. They said 'whoa', smiled and studied the cards thoroughly. They immediately caught the idea of the cards by looking at the figurative design and reading aloud, eager to tell about their experience: 'The builder is building, definitely', exclaimed the first student, continuing, 'We did a lot of building, adding papers, building a course for the "bot", Bitten [laughing]'. (Bitten was the name they gave the robot). The humorous expressions of the cards gave way to eager conversations about what happened. The card 'The imitator is imitating' made them wonder why they had not been looking around for inspiration (imitating) from the other groups. 'Whoa', one said. 'We usually look around because we get a bit bored'. Finally, they were asked to choose the most significant cards. They chose the 'fiddler', 'collector' and 'builder' cards. By selecting and studying the cards, they realised they had been 'all gone' (as they said) and totally absorbed (collecting, fiddling, building) in making 'Bitten's course' using the song 'YMCA'. In other words, the cards inspired specific insights concerning the situation's qualities of play. We also experienced a relaxed and humorous atmosphere during the interview.

In terms of the **reflection domain**, we learned from the second experiment that the cards shed light on the students' experienced learning situation in ways that the students found meaningful, and they were able to make connections between play qualities and the purpose of their education. They found it useful and appealing to explore the understanding

of their teaching practice and activities in which they were involved in using the cards. Second, we learned that the composition and figurative design of the cards engaged seemed intuitively meaningful to them, and we did not need to explain the specific meaning of the cards each time they had a conversation around the card. Third, we learned that the physical act of choosing and turning cards is playful, and turning cards, choosing cards, letting the students do the act of choosing in this case and turning engaged them in reflection on play qualities related to their education. In that sense, the use of the cards also became a playful experience for them in itself.

7. Discussion and conclusion

In this study, we explored how to develop a tool that supports playful learning in teacher education, both in the creation of playful learning solutions and learning about how to create playful learning solutions. The design considerations embedded in the Play Tarot Card prototype supported the cards' use through two design experiments with teachers and teacher students in teacher education. We illustrated how they can be used in exploring and understanding learning processes with play qualities, emphasising the diversity of play qualities. Furthermore, the experiments showed how they can inform design decision making by elaborating on existing learning designs and coming up with new ideas. The next step is to go further in developing the cards from a conversational tool to a hands-on design tool to be used at a specific phase of the design process, both for ideation and concept development to support play actions.

A question remains as to whether the play knowledge embedded in the tarot is at the right level of information for supporting teachers in their design process. In this paper, we underline the importance of focusing on actions as a gateway to play qualities. Bekker et al. (2014) argue about different levels and modalities of knowledge, but in this paper, it seemed challenging to focus on the action: During the tarot scenario, we had to keep motivating the participants to come back to the specific action. Having them stay with the action also pointed to the need to dwell on and explore previous experiences with designs and create a language for those experiences. The language of play qualities within teacher education also allowed them to share their experiences with their fellow colleagues and fellow students. The visuals of the cards worked well in terms of being open to interpretation. Lucero and Arrasvuori (2013) point to the importance of finding visuals for design cards that people can relate to and apply to diverse experiences, balancing abstraction and concrete figures. If the visuals are easy to relate to, it then becomes easier to interpret, thereby supporting openness.

For further development, there might be a need to be more explicit about the four main houses and explore how they can add more dimension to the tool—or if we should try to simplify it as much as possible, that the four categories help explain the theoretical framework for making the tool, but it is perhaps not necessary to have the same complexity when put into practice. By incorporating more information, the user of the cards can have additional layers to conceptualise experiences of play qualities. The user can also draw on

possible combinations of different houses and qualities that these represent or how they are being very strongly in one of these. Using these can also help in developing general instructions and descriptions of how to read the cards. Rather than having to give tight meanings to every single card, we could use the groups to help users become familiar with the highlighted qualities of that house. We will also consider adding helping questions or descriptions for each card through a guiding booklet for the reading facilitator. (We will not add such information to the cards because we want them to remain a flexible set of tarot cards).

At this point, the Play Tarot Cards have been used by us and the teachers together with the students. The next step is to explore how the Play Tarot Cards can be used by in-service teachers and their students.

Acknowledgements: We are grateful to the teachers and students for being involved in the Playful Learning Research Extension and for their contribution and involvement in the design experiments. We also thank LEGO Foundation for funding this research.

8. References

Auger, E. E. (2004). *Tarot and other meditation decks: History, theory, aesthetics, typology*. McFarland.

Bakhtin, M. (1981). *The dialogic imagination. Four Essays*. University of Texas Press.

Bakhtin, M. (1984). *Rabelais and his world*. Indiana University Press.

Barab, S., & Squire, K. (2004). Design-based research: Putting a stake in the ground. *The Journal of the Learning Sciences*, *13*(1), 1–14.

Bekker, T., & Antle, A. N. (2011). Developmentally situated design (DSD): Making theoretical knowledge accessible to designers of children's technology. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 2531–2540). ACM Press.

Bekker, T. & de Valk, L., & Eggen, J. H. (2014). A toolkit for designing playful interactions: The four lenses of play. *Journal of Ambient Intelligence and Smart Environments*, *6*(3), 263–276.

Boysen, M., Jensen, H., von Seelen, J. Jensen, H., J. Sørensen, M, & Skovbjerg, H. M. (2022, in press). Playful learning designs in teacher education and early childhood teacher education: A scoping review. *International Journal on Teaching and Teacher Education.*

Brown, A. L. (1992). Design experiments: Theoretical and methodological challenges in creating complex interventions in classroom settings. *The Journal of the Learning Sciences*, *2*(2), 141–178.

Charmaz, K. (2014). *Constructing grounded theory*. SAGE.

Christensen, O., Gynther, K., & Petersen, T. B. (2012). Tema 2: Design-based research – introduktion til en forskningsmetode i udvikling af nye E-læringskoncepter og didaktisk

design medieret af digitale teknologier. *Tidsskriftet Læring Og Medier* (LOM), *5*(9), p. 1-20.

Clarke, A. (2011). Situational analysis: Grounded theory mapping after the postmodern turn. *Symbolic Interaction*, *26*(4), 553–576.

Dewey, J. (1986). Experience and education. *Educational Forum*, *50*(3), 241–252.

Ejsing-Duun, S., & Skovbjerg, H. M. (2019). Design as modes of inquiry in design pedagogy. *International Journal of Art & Design Education*. https://doi.org/10.1111/jade.12214

Flick, U. (2000). Episodic interviewing. In M. W. Bauer & G. Gaskell (Eds.), *Qualitative researching with text, image and sound* (pp. 76–92). SAGE Publications. https://www.doi.org/10.4135/9781849209731

Hamaker-Zondag, K. (1997). *Tarot as a way of life: A Jungian approach to the Tarot*. Samuel Weiser.

Jensen, J.C., Lund, O., Pedersen, O., & Skovbjerg, H. M. (2021). Playful approaches to learning as a realm for the humanities in the culture of higher education: A hermeneutical literature review. *Arts and Humanities in Higher Education*, *0*(0), 1–22.

Jørgensen, H. H., Schrøder, V., & Skovbjerg, H. M. (2022). Playful learning, space and materiality: An integrative literature review. *Scandinavian Journal of Educational Research*.

Jørgensen, H. H., & Skovbjerg, H. M. (2020). "Det gjorde sygt, sygt ondt": design af indkluderende legemiljø i skolen i samarbejde med pædagoger. *Forskning i Pædagogers Profession og Uddannelse*, *4*(2), 109–121.

Lennon, K. (2015). *Imagination and the imaginary*. Routledge, Taylor & Francis Group.

Linley, J., Coulton, P., & Akmal, H. (2020). Turning philosophy with a speculative lathe: Object oriented ontology, carpentry, and design fiction. In *Proceedings of the Design Research Society Conference*.

Lucero, A., & Arrasvuori, J. (2013) The PLEX Cards and its techniques as sources of inspiration when designing for playfulness. *International Journal of Art and Technology*, *6*(1), 22–43.

Moseley, A., & Whitton, N. (Eds.). (2019). *Playful learning: Events and activities to engage adults.* Routledge.

Nørgård, R. T., Toft-Nielsen, C., & Whitton, N. (2017). Playful learning in higher education: Developing a signature pedagogy. *International Journal of Play*, *6*(3), 272–282.

Øksnes, M. & Sundsdal, E. (2020). *Barns lek i skolen*. Fagboklaget.

Pink, S. (2001). *Doing visual ethnography* (3rd ed.). SAGE Publications.

Pollack, R. (2002). *The Haindl tarot, the major arcana*. New Page Books.

Pramling, P., & Johansson, E. (2006). Play and learning—Inseparable dimensions in preschool practice. *Early Child Development and Care*, *176*(1), 47–65. https://doi.org/10.1080/0300443042000302654

Semetsky, I. (2011). *Re-symbolization of the self: Human development and tarot hermeneutic*. Springer.

Skovbjerg, H. M. (2020). Designing for play moods in a ludotorium. In Gudiksen, S. & Skovbjerg, H.M. *Framing play design: A hands-on guide for designers, learners and innovators* (pp. 37–49). BIS-Verlag.

Skovbjerg, H. M. (2021). *On play*. Samfundslitteratur.

Skovbjerg, H. M., Bekker, T., dÁnjou, B., Quinones, K.-Z.-P., & Johry, A. (2021). Examining theory use in design research on fantasy play. *International Journal of Child Computer Interaction*. https://doi.org/10.1016/j.ijcci.2021.100400

Skovbjerg, H. M., & Jørgensen, H. H. (2021). Legekvalitet - udviklingen af et begreb om det legende i lærer- og pædagoguddannelse. Læring og Medier.

Stenros, J. (2015) *Playfulness, play, and games. A constructionist ludology approach.* Tampere University Press.

Sutton-Smith, B. (2001). *The ambiguity of play*. Harvard University Press.

Whitton, N. (2018). Playful learning: Tools, techniques, and tactics. *Research in Learning Technology*, *26,* p. 1-12.

About the Authors:

Helle Marie Skovbjerg is Professor and Head of Research in LAB Design for Play, Design School Kolding, Denmark. Skovbjerg has been working on the mood perspective as a way to approach play, and has done a number of books and papers about that subject.

Helle Hovgaard Jørgensen has a Master in Comparative Literature, PhD and Associate Lecturer at The Department of Social Education, University College Lillebaelt, Denmark. Key research interests: Play in theory and practice, Media literacy, Narrativity an aesthetic theories, Child and Youth Culture.

Keila Zari Pérez Quiñones is a play–designer working with social–politics for social change and innovation through play and playfulness. She teaches, creates and contributes as a research assistant at the LAB for Design & Play, Design School Kolding, Denmark.

Tilde Bekker is Professor at the Industrial Design department of Eindhoven University of Technology. She is a design researcher working on theory-inspired design for play and learning. She developed various design methods and design tools and has published over 110 papers.